AGENTIC AI (26CD7PEAAI) · MINI-PROJECT REVIEW 2: DESIGN & ARCHITECTURE

Ariadne

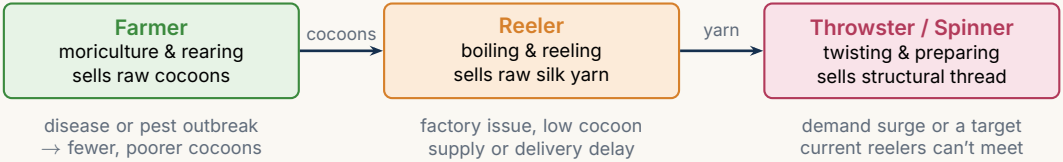
Sericulture supply-chain resilience through agentic triage and dynamic rerouting

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CONTEXT

One disruption, three tiers



Role-aware intake

Every actor reports an incident or request through one chatbot; each report becomes a tracked incident.

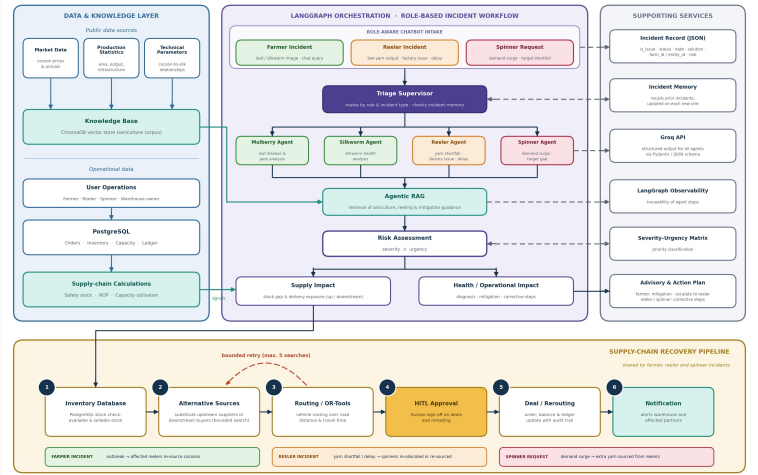
Triage, then recover

Disease triage, grounded retrieval and a severity–urgency rating, then inventory-aware rerouting up- or downstream.

A human signs every deal

No order, balance or ledger change happens without explicit approval.

System architecture



Three cooperating layers

1 Data & knowledge — public sources → ChromaDB; operational data in PostgreSQL feeding supply-chain calculations.

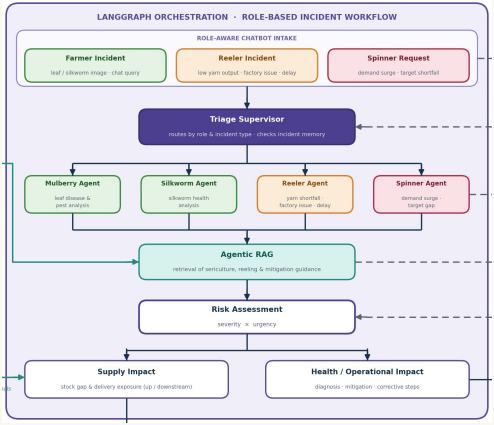
2 LangGraph orchestration — one supervisor, four specialists, Agentic RAG and risk assessment.

3 Recovery pipeline — six stages shared by all three entry points.

Supporting services span all layers: incident record & memory, Groq structured output, tracing, severity–urgency matrix.

Solid: control/data flow. Dashed: supporting-service calls. Green, amber and rose mark the farmer, reeler and spinner paths.

Inside the agent graph



Triage Supervisor

Checks incident memory, then routes by role and incident type. Every specialist handoff passes through it.

Four specialist agents

Mulberry leaf disease & pests · **Silkworm** health · **Reeler** yarn shortfall, factory issue, delay (stock, degradation time, capacity) · **Spinner** demand surge, target gap.

Agentic RAG → Risk Assessment

Guidance retrieved from ChromaDB; the incident is rated on the severity–urgency matrix and split into a **health/operational** branch (advisory & action plan) and a **supply-impact** branch.

Data flow: from report to approved reroute

1 Ingestion

Image, report or request creates an incident stored as JSON: `is_issue`, status, date, solution, entity id, role.

→

2 Triage & retrieval

Supervisor checks incident memory and hands off; the specialist queries Agentic RAG.

→

3 Risk assessment

Severity × urgency, then split into a health/operational advisory and a supply impact.

→

4 Supply impact

Stock, safety stock, ROP and capacity from PostgreSQL: who is affected, how much stock is at risk.

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5 Recovery

Bounded search (≤ 5) for alternatives in the needed direction; OR-Tools plans the routes.

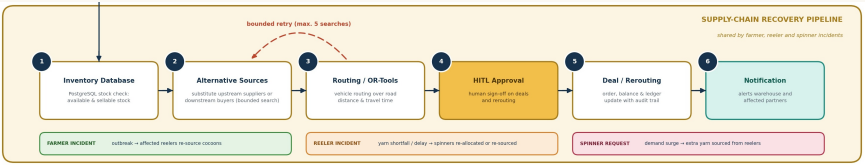
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6 Approval & execution

HITL gate, then ledger update and notification of the warehouse and partners.

Cascades are linked, not lost. A farmer outbreak means fewer cocoons; the reeler may then report a yarn shortfall, opening a reeler incident for downstream spinners. Incident memory links the two so the second triage starts with the first one's context.

One recovery pipeline, three entry points



Entry point	Trigger	Specialist & assessment	Reroute & deal
Farmer	Leaf/silkworm image or a chat query on a suspected disease or pest	Mulberry / Silkworm agent + RAG: diagnosis, severity, urgency, mitigation, escalate?	Affected reelers found; alternative farmers searched if cocoon supply at risk; farmer–reeler deal
Reeler	Too little yarn (factory issue or low cocoons) or a delivery delay	Reeler agent + RAG + stock: severity, urgency, time to degradation, options	Affected spinners found; yarn re-allocated or alternatives sourced; reeler–spinner deal
Spinner	Demand surge, or a target current reelers may miss	Spinner agent + safety-stock & capacity: gap vs. target, urgency	Alternative reelers searched and routed for the shortfall; reeler–spinner deal

Design controls

Bounded loops
Alternative-source search stops after at most 5 attempts; if nothing fits, the workflow escalates instead of looping.

Role-based visibility
Each user sees only his own tier: a mulberry farmer his farm; a silkworm farmer his supplier and his farm; a warehouse owner his linked farmers and warehouse.

Incident memory
Recurring issues recall earlier incidents; memory is updated after every new one.

Observability
LangGraph traces every agent step, so routing, tool calls and the termination point can be audited and scored.

Structured outputs
Groq API responses validated against Pydantic/JSON schemas: machine-checkable, not free text.

Human oversight
HITL approval finalises every farmer–reeler, reeler–spinner and farmer–farmer deal before the ledger changes.

Patterns chosen, and why

Pattern	Where it is applied	Why this problem needs it
Multi-agent with supervisor	Triage Supervisor coordinating Mulberry, Silkworm, Reeler and Spinner agents; all collaboration supervisor-mediated	Farm, factory and demand problems need distinct expertise; a central supervisor keeps routing and handoffs controlled
Agentic RAG	Agents query ChromaDB: disease management, rearing & reeling practice, mitigation, warehouse documents	Diagnoses and resolutions must rest on documented guidance, not model memory
Planning & rerouting	Risk → Supply impact → Inventory → Alternative sources → Routing, shared by all entry points, bounded at 5 searches	Recovery is a staged plan whether the shortfall is upstream (cocoons) or downstream (yarn); the bound prevents runaway iteration
Tool use	Groq structured output, PostgreSQL inventory calls, OR-Tools paths and distances	Stock figures and routes must come from authoritative tools, not generated text
Human-in-the-loop	Approval gate before every deal or reroute	Deals change balances and stock, so a human decides before the ledger is updated
Memory & observability	Incident memory read on recurrence, written after each incident; LangGraph tracing	Repeated issues reuse context; decisions stay debuggable and auditable

Alternatives we rejected

× A single monolithic agent

- Mixes farm pathology, factory operations and demand planning in one prompt and one tool set.
- A wrong route is hard to trace: no handoff to inspect, no routing accuracy to measure.

Chosen instead: supervisor + four specialists, each with a narrow tool set and its own schema.

× Fully autonomous deals

- One wrong inference — a misdiagnosis, a stale stock figure — would move money and stock directly.
- Deals cross organisations: farmers, reelers and spinners must consent.

Chosen instead: the agent prepares the deal and route; a human approves before any ledger change.

Data plan: knowledge and vision

#	Data required	Source	Intended use
1	Mulberry disease & pest images	PaddlePaddle dataset (6 classes, 1,788 original images)	Train / fine-tune the mulberry vision model
2	Silkworm disease images	Silkworm disease research dataset	Healthy-vs-diseased classifier CONFIRM DOWNLOAD
3	Disease treatment & sericulture knowledge	Central Silk Board; SILKS; Silk India	ChromaDB corpus: disease management, rearing, advisories, mitigation
4	Sericulture statistics & technical knowledge	International Sericultural Commission	Mulberry, silkworm breeds, industry statistics
5	Cocoon quality, reeling yield, capacity, storage	FAO Silk Reeling and Testing Manual	Cocoon-to-silk requirements and processing constraints
6	Farmers, mulberry area, production, reeling infrastructure	Ramanagara District Sericulture Department	A real regional network and its approximate scale

Data plan: network and operations

#	Data required	Source	Intended use
7	Cocoon market locations & procedures	Karnataka SILKS marketing portal	Market nodes and farmer-to-reeler procurement paths
8	Historical cocoon prices & arrivals	*Price Dynamics of Mulberry Cocoons in South India*	Price trends, supply variation (numeric records if available)
9	Farm, reeler, spinner, warehouse locations	OpenStreetMap + registered user locations	Real distances; no public registry of private participants exists
10	Road distance & travel time	Google Routes API or OSM routing	Distance / time matrix for OR-Tools
11	Inventory, sellable & safety stock	User entries, stock receipts, approved transactions	Authoritative balances; ROP and safety stock from observed demand & lead times
12	Deals, balances, incident history	Application transactions & incident reports	HITL approvals, ledger, incident memory, audit trail

Supply-chain calculations feeding the agents

Reorder point and safety stock

$$\text{ROP} = \bar{d}\,L + SS \qquad SS = z\sqrt{L\,\sigma_d^2 + \bar{d}^{\,2}\sigma_L^2}$$

$\bar{d}$: mean daily demand; L: lead time; σ_d, σ_L: their standard deviations; z: service-level factor.

When an incident forces a farther source, L grows, ROP rises, and the Reeler or Spinner agent sees the gap before stock runs out.

Capacity utilisation and cocoon-to-yarn

$$U = \frac{\text{actual output}}{\text{rated capacity}} \qquad \text{yarn} \approx \frac{\text{cocoons}}{\text{renditta}}$$

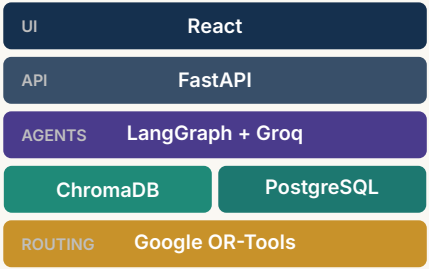
Renditta: kg of cocoons needed per kg of raw silk.

A farmer incident lowers cocoon quantity *and* quality (higher renditta), so the reeler's yarn shortfall is larger than the weight drop alone suggests.

Inputs come from the application's own transactions (data source 11) and FAO / CSB references (sources 3, 5); the LLM never produces these numbers.

Tool plan and technology stack

Tool	Role in the graph
Groq API	Structured output (free tier), validated against Pydantic/JSON schemas
ChromaDB retrieval	Agentic RAG over sericulture and warehouse documents
PostgreSQL calls	Inventory checks; orders, stock, capacity and ledger records
OR-Tools + Routes API / OSM	Vehicle routing over road distance and travel time
LangGraph	Supervisor, specialists, bounded loops; observability



How quality will be judged: four dimensions

A Workflow & tools

Does the graph do the right thing?

- Task completion
- Routing accuracy
- Tool selection
- Output validity
- Tool execution
- Handoff accuracy
- Unnecessary calls
- Termination accuracy

B Answer quality

Is the answer right and useful?

- Correctness
- Completeness
- Relevance

C Grounding

Does it rest on what was retrieved?

- Groundedness / faithfulness
- Citation accuracy
- Recall@K
- Context precision@K

D Uncertainty

Does it know when to stop?

- Abstention accuracy
- Hallucination rate

17 metrics, each computed over a test set covering all three entry points: farmer incidents, reeler incidents and spinner requests.

Metrics A: agent workflow and tool behaviour

Metric	Computation	Success means
Task completion rate	completed tasks ÷ total test tasks	Intended workflow completed
Supervisor routing accuracy	correctly routed ÷ routing cases	Right specialist reached
Tool selection accuracy	correct tool calls ÷ tool-selection cases	Right DB, RAG or routing tool
Structured output validity	schema-valid outputs ÷ outputs tested	Conforms to Pydantic/JSON
Tool execution success	successful ÷ attempted valid executions	Calls succeed when preconditions hold
Handoff accuracy	correct handoffs ÷ handoffs evaluated	Context arrives intact
Unnecessary tool-call rate	unnecessary calls ÷ total calls (lower is better)	No redundant calls or loops
Termination accuracy	correctly completed / rejected / escalated ÷ tested	Graph ends or pauses at the right point

All ratios reported as percentages. Scored automatically from LangGraph traces against the expected route, tool calls and termination point.

Metrics B–D: answers, grounding, uncertainty

	Metric	Computation	Success means
B	**Answer correctness**	correct answers ÷ factual questions	Agrees with verified facts
	Completeness	judge score vs. checklist of required elements	Covers the important parts
	Relevance	LLM-judge score of directness	No irrelevant content
C	**Groundedness**	supported claims ÷ claims checked	No unsupported claims
	Citation accuracy	supporting citations ÷ citations evaluated	Citations back their claims
	Retrieval Recall@K	relevant in top *K* ÷ all known relevant	Retrieves the right material
	Context Precision@K	relevant in top *K* ÷ all in top *K*	Little irrelevant context
D	**Abstention accuracy**	correct abstentions/escalations ÷ cases needing them	Admits uncertainty, asks a human
	Hallucination rate	unsupported factual claims ÷ claims evaluated	Minimised

How the evaluation is run

1. **Test sets** — scripted scenarios for each entry point, plus edge cases.
2. **Workflow (A)** — scored from LangGraph traces: recorded routing, tool calls and termination vs. expected.
3. **Answers (B)** — verified reference answers and a checklist-based LLM judge.
4. **Grounding (C)** — each claim and citation checked against retrieved chunks; retrieval scored against hand-labelled relevant documents.
5. **Uncertainty (D)** — cases built with insufficient evidence, where the right move is to abstain or escalate.

Thresholds are fixed after a first baseline run, not guessed up front.

Edge cases in the test set

- Zero stock at the reporting node
- No alternative source within the 5-search bound
- A deal rejected at the HITL gate
- A question the knowledge base cannot answer
- A farmer incident that cascades into a reeler incident

Thank you

Questions and feedback welcome

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